

AMENDMENT NO. 1 JUNE 2024

TO

IS 6595 (PART 1) : 2018 HORIZONTAL CENTRIFUGAL PUMPS FOR CLEAR, COLD WATER — SPECIFICATION

PART 1 AGRICULTURAL AND RURAL WATER SUPPLY PURPOSES

(*Fourth Revision*)

(*Second cover, foreword, sixth paragraph*) — Insert the following at the end:

‘Graphs for calculating minimum pump efficiency are replaced by equation (1). Equation (1) is adopted from Commission Regulation (EU) No. 547/2012 with regarding to eco-design requirements for water pumps for minimum efficiency index (MEI), for pump type end suction own bearing (ESOB).’

(*Page 2, clause 8.2*) — Substitute the following for the existing:

‘**8.2** The minimum pump efficiency at the specified duty point shall be in accordance with efficiency calculated as per **8.4** using C factor of 4 pole speed from 1 200 to 2 000 rpm and using C factor of 2 pole speed from 2 001 to 3 600 rpm.’

(*Page 2, clause 8.3*) — Insert the following new clause at the end:

‘**8.4 Pump Efficiency**

Pump efficiency shall be calculated using the equation as given below, and efficiency corresponding to MEL 0.2 shall be taken as the minimum pump efficiency:

$$\eta_p = 88.59 x + 13.46 y - 11.48 x^2 - 0.85 y^2 - 0.38 xy - C \dots\dots\dots (1)$$

where

η_p = minimum pump efficiency;
 $\ln$ = natural logarithm;
 y = $\ln(Q)$;
 x = $\ln(n_s)$;
 n_s = specific speed; and
 Q = discharge rate, in m³/h.

$$n_s = n \cdot \frac{\sqrt{Q/3\,600}}{(H)^{0.75}}$$

where

n = rotational speed, in revolutions per minute (rpm);
 H = head, per stage in meters m;
 Q = discharge rate, in m³/h; and
 C = constant for pump type and different MEL levels.

Price Group 2

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Minimum Efficiency Level, MEL		0.20	0.30	0.40	0.50	0.60	0.70
C	2 Pole, speed	133.43	131.61	130.27	129.18	128.12	127.06
	4 Pole, speed	130.68	129.35	128.07	126.97	126.10	124.85

NOTES

- 1 Other MEL values are for the guidance of the manufacturer and to upgrade to higher values of pump efficiency.
- 2 An illustrative example for calculating pump efficiency using above equation is given in Annex B.
- 3 The calculated value for specific speed and minimum pump efficiency shall be rounded off up to 2 decimal places.

(Page 5, clause 12.2, line 1) — Substitute ‘The efficiency of the pump shall not be less than those values derived from equation (1) of 8.4.’ for ‘The efficiency of the pump shall not be less than those values given in Fig. 1 to Fig. 4.’

[Page 5, clause 12.2.3, a), line 3] — Substitute ‘However, after applying the tolerance, efficiency value shall not be less than that obtained from equation (1) of 8.4.’ for ‘However, after applying the tolerance, efficiency value shall not be less than that derived from Fig. 1 to Fig. 4.’

(Page 3, Fig. 1) — Delete.

(Page 4, Fig. 2) — Delete.

(Page 6, Fig. 3) — Delete.

(Page 7, Fig. 4) — Delete.

(Page 9, Annex A) — Insert the following at the end:

‘ANNEX B

(Clause 8.4)

**EXAMPLE OF MINIMUM PUMP EFFICIENCY CALCULATION FOR HORIZONTAL
END SUCTION VOLUTE TYPE CENTRIFUGAL PUMP (4 POLE) DUTY POINT**

B-1 INPUT DATA

Head = 10 m
Discharge rate = 16 lps
Speed = 1 440 rpm

B-2 CALCULATIONS

B-2.1 To Find out Discharge Rate, Q, in m³/h

Discharge rate = 16 lps × 3.6 = 57.6 m³/h.

B-2.2 To Find out Specific Speed, n_s

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

$$n_s = 1440 \times \frac{\sqrt{57.6/3600}}{(10)^{0.75}} = 32.39$$

NOTE — For specific speed calculation we have to use discharge (Q), in m³/h.

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B-2.3 To Find out Minimum Pump Efficiency, η_p

$$\eta_p = 88.59 x + 13.46 y - 11.48 x^2 - 0.85 y^2 - 0.38 xy - C \dots \dots \dots (1)$$

$$x = \ln(n_s) = \ln(32.39) = 3.47$$

$$y = \ln(Q) = \ln(57.6) = 4.05$$

NOTE — For minimum pump efficiency calculation we have to use discharge (Q), in m³/h.

C = 130.68 (C-value for MEL 0.2 to be taken from the table for minimum efficiency calculation for 4 pole)

$$\eta_p = 88.59 \times 3.47 + 13.46 \times 4.05 - 11.48 \times 3.47^2 - 0.85 \times 4.05^2 - 0.38 \times 3.47 \times 4.05 - 130.68$$

$$\eta_p = 73.70 \% \text{ (Minimum pump efficiency) '}$$

(MED 20)

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